

Fig 12.9 High temperature grilling of meat produces polycyclic aromatic hydrocarbons

- Fats and oils can become oxidized on exposure to light, air or heat. Many of the oxidation products are carcinogenic. Storing oil in airtight containers prevents the oil from getting oxidized.
- Avoid intake of processed meat.
- Avoid ingestion of foods that look or smell mouldy, as they may contain toxic carcinogens like aflatoxin.
- The inclusion of fresh organic fruits and vegetables containing different natural plant pigments in the daily diet ensures meeting the phytochemical and antioxidant needs.
- Liberal intake of water prevents accumulation of toxic wastes by flushing them out of the body.
- A healthy diet, adequate physical activity and maintaining an ideal body weight reduce cancer risk.
- Excess intake of calories in the form of sugar and fat lead to excess body fat that has been strongly linked to several types of cancers.
- Trans fats are formed when liquid vegetable oils are made solid by the process of called hydrogenation. These fats are commonly used in processed food such as cookies, crackers, snack foods, fried foods, and pastries as they give foods the desirable taste, shape, and texture. Research has shown that regular intake of foods rich in trans-fat increases the risk for cancer.
- Smoking and tobacco chewing are the most preventable factors in the incidence of oral and lung cancer.
- Excessive intake of alcohol may act as an irritant, especially in the mouth and throat. Cells that are damaged by alcohol may undergo alteration in their DNA leading to cancer.
- Avoid exposure to harmful radiation and chemicals that cause cancer.

SUMMARY

- ❖ Cancer is a disease characterized by abnormal cell division. It can occur in any body tissue.
- ❖ Energy and nutrients needs increase during cancer as the tumour and normal cells compete with each other for their supply of nutrients. This hypermetabolic state coupled with a lack of appetite and poor intake of food leads to extreme muscle wasting termed cancer cachexia.
- ❖ Treatment of cancer includes surgery, radiation, and chemotherapy.
- ❖ Improving the patient's nutritional status is a challenge due to the aggressive nature of the illness and anorexia. In such a scenario, resorting to parenteral or enteral nutrition may be necessary.

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